



BIP-39 seed phrase  
(12/15/18/21/24 words,  
10 supported wordlists)

entropy bytes  
16/20/24/28/32 B  
(deterministic wordlist  
lookup)

Payload::Entr(bytes)  
+ 0x00 prefix

envelope::package  
→ rust-codex32::from\_seed  
(HRP=ms, threshold=0,  
id=entr,  
share=s, payload, short  
BCH)

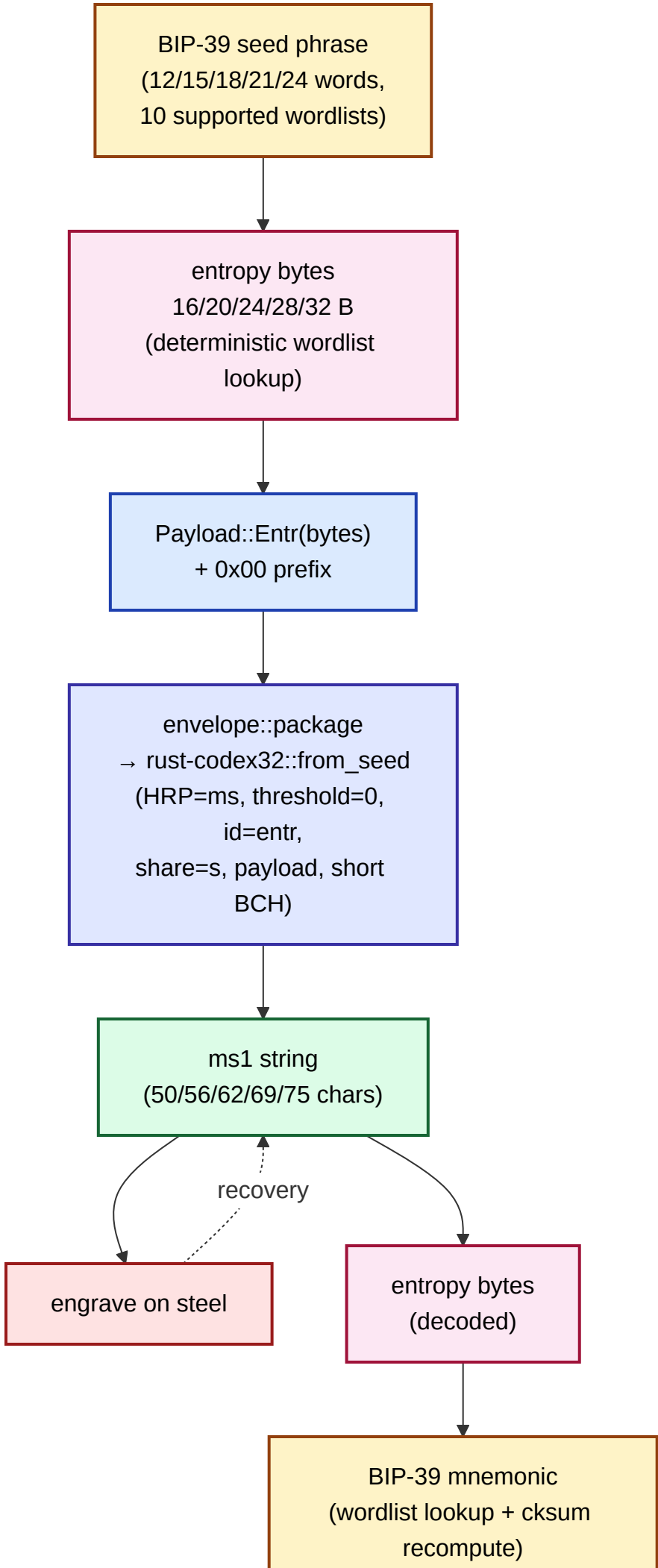
ms1 string  
(50/56/62/69/75 chars)

engrave on steel

recovery

entropy bytes  
(decoded)

BIP-39 mnemonic  
(wordlist lookup + cksum  
recompute)



```
graph TD; A[ ] --> B["BIP-32 master seed (64 B)  
via PBKDF2-HMAC-SHA512  
(mnemonic + passphrase,  
2048 iters)"]; style A fill:#ffffcc,stroke:#8b4513,stroke-width:1px; style B fill:#ffffcc,stroke:#8b4513,stroke-width:1px;
```

BIP-32 master seed (64 B)  
via PBKDF2-HMAC-  
SHA512  
(mnemonic + passphrase,  
2048 iters)